Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
B		2630979

The Space Elevator Imperative: A Multi-Dimensional Assessment of Lunar Logistics under Planetary Environmental Constraints

Summary

Efficient, economical, and sustainable logistics are the core challenge for establishing a human Moon colony. This paper aims to systematically evaluate and compare three logistics scenarios—the Space Elevator System, traditional rocket transport, and a hybrid strategy of both—by constructing a series of mathematical models, providing strategic analysis and optimization recommendations.

We established a mathematical analysis workflow composed of four core problems:

For Problem 1—evaluating and comparing the cost and time of the scenarios, this paper builds a **spatiotemporal logistics transport model**. This model includes a space elevator capacity sub-model and a rocket capacity sub-model, combined with an economic optimization model based on opportunity cost. Through **deterministic simulation**, we found that the option of Rockets Only would take about **285 years** and incur an opportunity cost of **$8.5 trillion**, while the SES-Only option would take only **37.5 years**.

For Problem 2—exploring system performance under non-ideal conditions, we developed a **"soft degradation" reliability model for the space elevator** and a **"hard failure" vulnerability model for rockets**, integrating them into an extended stochastic reliability and robustness analysis model. We used the **Monte Carlo simulation** method to evaluate the robustness of the transportation system.

For Problem 3—addressing the water supply needs for the first year after the colony is established, we constructed a **net supply gap assessment model based on supply-demand balance**, a **multi-modal transport model**, and a **logistics balance model** for inventory and time dynamics. By solving the equilibrium equation, we calculated a water cycle efficiency threshold of **73.25**% and determined the transportation plan to ensure the colony's water security.

For Problem 4—analyzing the environmental impact of different transport scenarios, we established a **planetary-scale environmental impact assessment model**. This model integrates sub-models for stratospheric water vapor, orbital debris, and the Kessler Syndrome, quantifying the pollution and space debris risks from launch activities. The results indicate that the rocket strategy would trigger an uncontrollable Kessler syndrome within just **5 years** of operation, making it physically unfeasible.

In summary, by integrating cost, risk, long-term support, and environmental impact, we conclude that **the SES-Only scenario is the optimal choice** to achieve the goal, as it offers zero emissions, low cost, and high reliability. Meanwhile, we recommend retaining rockets as a high-capacity emergency backup.

Finally, we conducted a **sensitivity analysis** on the model's key parameters to test the robustness and limitations of the scenarios. We also evaluated the strengths of our models to provide a solid analytical basis for our final recommendations.

Keywords: **Time-varying spatiotemporal model, Multi-objective optimization, Monte Carlo simulation, Segmented priority scheduling, Kessler effect model**

Contents

1 Introduction

1.1 Problem Background

With humanity's ever-growing desire for deep space exploration and increasing concern over Earth's ecological pressures, the construction of large-scale off-world colonies has transitioned from science-fiction concepts to strategic planning. Humanity plans to initiate a 100,000-person Moon Colony project in 2050, facing the logistical challenge of transporting up to 100 million metric tons of construction and supply materials from Earth to the Moon.

To address the challenge of Earth-Moon transportation, a Space Elevator System has been designed. Powered by electricity, it can lift vast quantities of materials from the Earth's surface to geosynchronous orbit (GEO) and beyond in an environmentally friendly manner, significantly reducing fuel consumption. As another route, traditional rockets are also continuously advancing. Through technological iterations, their payload capacity per launch continues to increase, enabling them to deliver materials directly from Earth-based launch sites to the Moon Colony.

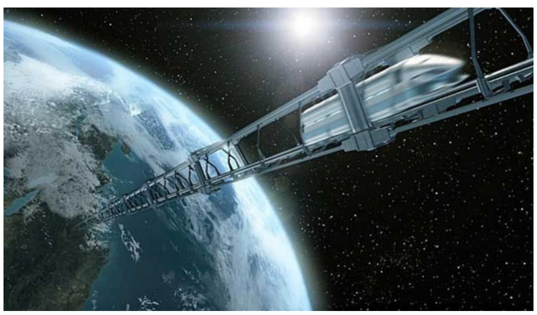

Figure 1: Space Elevator System

Figure 2: Traditional Rockets

The two approaches differ in aspects such as transportation modes, cost structures, suggesting that a single technological path may not be the optimal solution. Therefore, the key to accomplishing this grand interplanetary "relocation" mission is to find an optimal strategy for the construction and subsequent sustainable operation of the Moon Colony project, one that balances efficiency, cost-effectiveness, and sustainability.

1.2 Restatement of the Problem

To solve the planning problem of transporting construction materials and supplies to the Moon Colony, we have developed a comprehensive mathematical model. This model is designed to quantitatively evaluate and optimize different transportation strategies and systematically answer the following four core questions:

- Problem I: Evaluate and compare the total cost, timeline, and feasibility of three core transportation strategies (a. a Space Elevator-only strategy, b. a rocket-only strategy, and c. a hybrid strategy) for completing the specified transportation mission.
- Problem II: Analyze what adjustments would be needed for our recommended optimal strategy when the transportation system experiences non-ideal conditions, and quantify the impact of these adjustments on the total project cost and duration.

- Problem III: Apply our transportation model to analyze the transport tasks required to ensure a sufficient year-round water supply for the Moon Colony after it enters full operational phase, and calculate the associated additional costs and timeframes.
- Problem IV:Analyze the environmental impact on Earth of establishing the 100,000-person Moon Colony under different transportation scenarios, and explore how to adjust our model's strategy to achieve minimization of this environmental impact.

1.3 Our Work

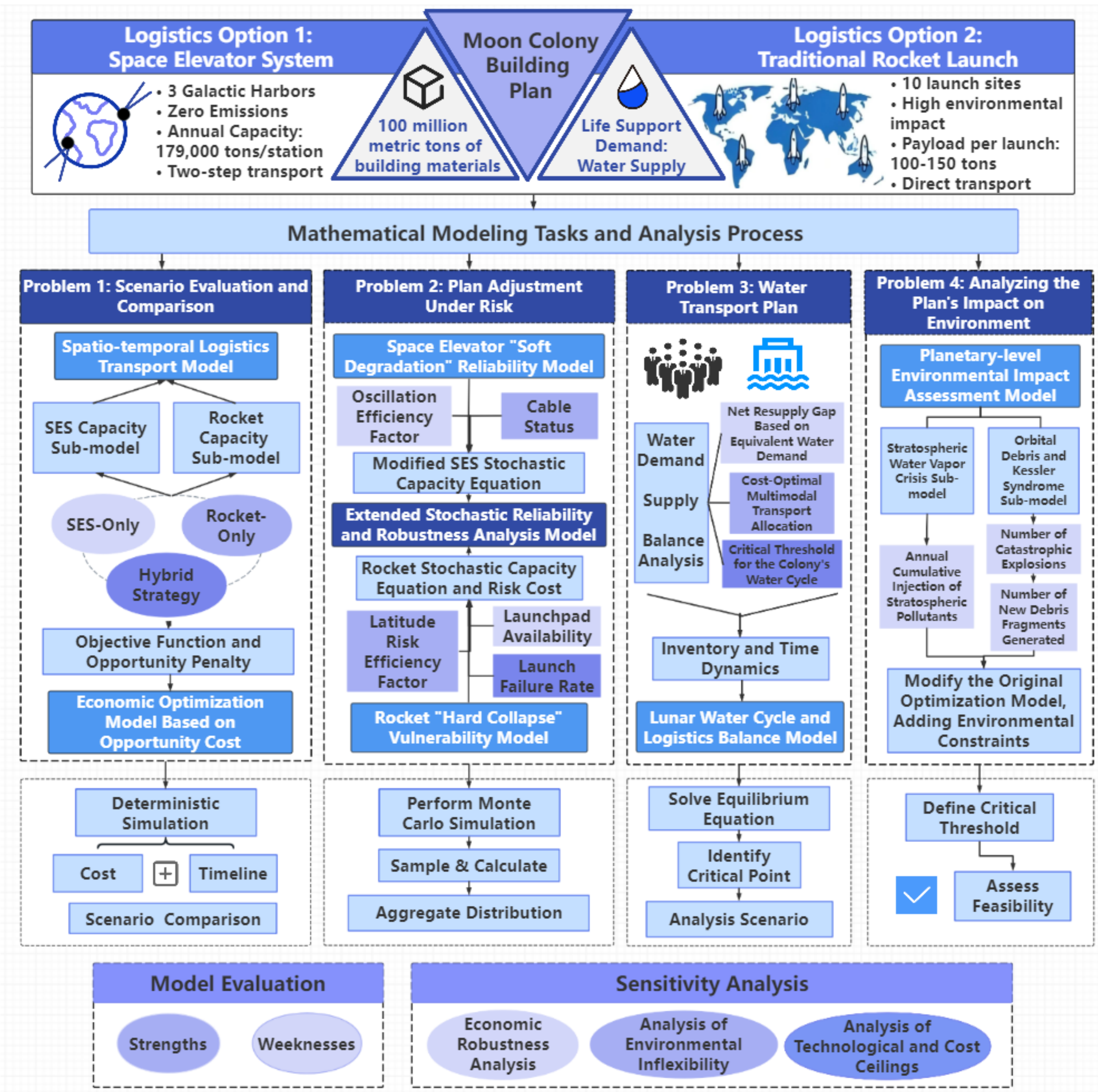

Figure 3: Our Work

2 Model Preparation

2.1 Assumptions and Justifications

- **Assumption 1**: The model does not consider building new rocket launch sites but instead considers an increase in maximum launch frequency.

  Justification :The main drivers for rocket development come from the maturation of reusable technology and the automation of ground support procedures.

- **Assumption 2**: The transfer time from the space elevator's apex anchor to the Moon is considered negligible.

  Justification : Including this minor delay in the macroscopic dynamic model would add unnecessary complexity without substantially affecting the project's long-term strategic results.

- **Assumption 3**:The lunar colony's population remains stable during the one-year period being analyzed, and short-term population fluctuations are ignored.

  Justification :Treating the population as a constant simplifies the problem from a complex population dynamics issue into a purely logistical one.

- **Assumption 4**:The average daily water requirement for all individuals on the lunar colony is uniform, set at 50kg/day.

  Justification : This avoids the need to create complex consumption models for different population segments, which simplifies the calculations.

- **Assumption 5**:The mass of the containers used for transporting water is considered negligible.

  Justification : This reduces the model's complexity by avoiding additional parameters, without altering the core conclusions.

- **Assumption 6**: All heavy-lift launch vehicles use liquid oxygen-methane fuel, which does not produce black carbon.

  Justification : This aligns with the technological trend for next-generation rockets and allows the model to focus on the more critical issue of stratospheric water vapor.

- **Assumption 7**: The average residence time of water vapor injected into the stratosphere is 3-5 years.

  Justification :The stratosphere lacks effective precipitation-based removal mechanisms, leading to the long-term retention of pollutants and causing a greater impact.

- **Assumption 8**:The generation rate of new debris from collisions is proportional to the square of the total number of debris fragments.

  Justification : This is based on the physical model of the Kessler Syndrome and serves as the mathematical foundation for the exponential growth in debris population.

- **Assumption 9**:During the simulation period, no technology exists that can clear vast amounts of small orbital debris instantaneously.

Justification :This ensures the model only considers the natural decay of debris based on existing physical laws, rather than relying on science-fiction technologies.

2.2 Notations

Table 1: Model Variables by Type, with Definitions and Typical Values

Type	Symbol	Definition	Typical Value / Unit
Global Variable	t	Time	Year
	T	Time Required For Project	Year
	Q_{total}	Total Demand	10^8 tons
	r_{disc}	Discount Rate	3%
Water Resource	P	Population	100,000
	ω	Per Capita Water Demand	0.05 tons/day
	α	Recycling Rate	98%
	D	Initial Safety Stock Days	1000 days
Logistics	$K_{\text{SES}}(t)$	Space Elevator Annual Capacity	tons/year
	$K_{\text{Rocket}}(t)$	Rocket System Annual Capacity	tons/year
	η_{lat}	Latitude Efficiency Factor	$0.6 \sim 1.0$

Note: There are some variables that are not listed here and will be discussed in detail in each section.

3 Models Establishment and Solution of Problem 1

3.1 Model I : Dynamic Logistics Transport Model

To accurately describe the engineering of Earth-Moon transport, we have moved beyond the limitations of traditional methods that use static capacity. We believe that the effective capacity of both the emerging Space Elevator and mature rocket technology will evolve over time. Therefore, we have developed a "Time-Varying Spatiotemporal Logistics Model." This model not only considers the non-linear changes in system capacity over time but also incorporates key physical and geographical constraints, providing a more solid foundation for project timeline prediction .

The core idea of this model is to treat the transport of materials between Earth and Moon as a dynamic accumulation process[1]. Its accumulation rate depends on the combined, time-varying capacity of the different transport tools. Based on our model assumptions, we express the annual transport capacity of the two major systems mathematically:

3.1.1 Space Elevator System Capacity Sub-model

We view the SES as a set of parallel, continuous material conveyor belts[1]. Its instantaneous annualized capacity $K_{SES}(t)$ at any given time t is defined by the following basic formula:

$$K_{SES}(t) = N_{SES} \cdot M_{load} \cdot f_{freq}(t) \cdot \tau_{op} \tag{1}$$

- N_{sys} is the system scale, the total number of tethers. We adopt a maximized ascent strategy—during launch windows, all 6 tethers from the 3 Galactic Ports are used for uplink cargo to maximize hardware utilization, so $N_{sys} = 6$.

- M_{load} is the effective payload of a single climber. Referencing the International Space Elevator Consortium's FOC standard, we set the standard payload for a single climber at 20 tons.
- τ_{op} is the annual operating hours. We assume year-round, non-stop operation, totaling 24 hours/day $\times$ 365 days/year $= 8760$ hours.

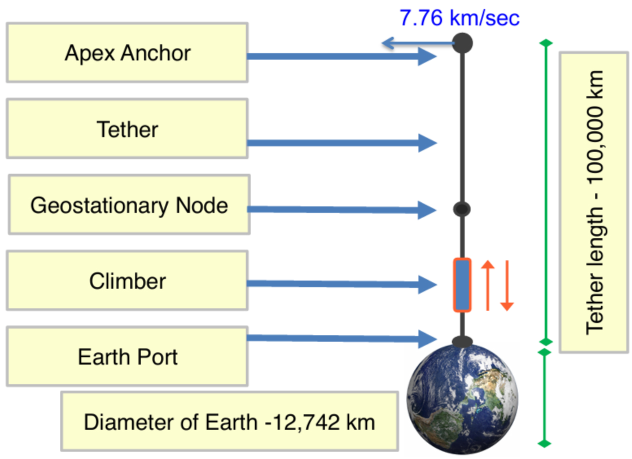

Figure 4: Space Elevator Structure Diagram

The most critical variable in the formula is the departure frequency $f_{freq}(t)$, which determines the system's turnaround efficiency. We innovatively propose that this frequency is primarily constrained by two core technical parameters : climber speed and safe separation distance:

$$f_{freq}(t) = \frac{V(t)}{H(t)} \tag{2}$$

The climber's propulsion technology requires a maturation process[2]. Its average operating speed will go from a slow start to rapid improvement, eventually approaching the design limit. We use a logistic growth model to describe this process, with the average climber speed $V(t)$ in year t expressed as:

$$V(t) = \frac{V_{max}}{1 + e^{-k_v(t - t_{mid_v})}} \tag{3}$$

- V_{max} represents the maximum speed after the technology is fully mature. We set the maximum speed $V_{max} = 3000$ km/h.
- $t_{mid_v} = 2075$, the year when the speed reaches its growth inflection point. We delay the technology maturation period to simulate the slow ramp-up of a large-scale project;
- the growth rate coefficient $k_v = 0.15$, meaning the main phase of technology ramp-up is about 15-20 years.

Similarly, a large safety distance must be maintained between departing climbers to avoid tether wobble and collision risks.[2][3] As advanced AI attitude control and coordinated scheduling technologies mature, this distance can be significantly reduced. We use a reverse logistic function to describe this optimization process:

$$H(t) = H_{min} + \frac{H_{max} - H_{min}}{1 + e^{k_h(t - t_{mid\_h})}} \tag{4}$$

To ensure the model strictly conforms to the information given in the problem, we used a "reverse calibration method". The problem gives the annual capacity of a single Galactic Port in 2050 as 179,000 tons. Based on this target, and under the physical constraints of $V_{min} = 500$ km/h and $M_{load} = 20$ tons, we back-calculated the initial safe separation distance H_{max}:

$$H_{max} = \frac{\text{Tethers per Port} \times M_{load} \times V_{min} \times \tau_{op}}{\text{Target Capacity per Port}} = \frac{2 \times 20 \times 500 \times 8760}{179,000} \approx 979 \text{ km} \tag{5}$$

This indicates that in the project's initial phase, a large safety distance of approximately 979 km must be maintained between climbers. From the calibration, we get an initial distance $H_{max} = 979$ km, which eventually converges to $H_{min} = 50$ km . $t_{mid\_h} = 2080$, indicating control technology matures slightly later than velocity technology; $k_h = 0.2$.

After integration, the total annual capacity formula for the Space Elevator is:

$$K_{SES}(t) = 6 \times 20 \times \left(\frac{V(t)}{H(t)}\right) \times 8760 \tag{6}$$

3.1.2 Traditional Rocket Capacity Sub-model

Unlike the elevator, rocket transport is discrete and pulse-like. The core of its capacity model is the geospatial differences. Launch sites at different geographical latitudes have varying launch efficiencies due to differences in the Earth's rotational speed. A launch site near the equator can better utilize the Earth's linear velocity, thus consuming less fuel to launch the same payload mass into a lunar transfer orbit, giving it higher efficiency.

To account for this, we model 10 launch sites, assigning each site an efficiency coefficient $\eta_{\text{lat}} \in [0, 1]$. We then take a weighted sum of the capacities of these 10 launch bases. The total annual capacity $K_{Rocket}^{total}(t)$ is defined as:

$$K_{Rocket}^{total}(t) = M_{base} \cdot \sum_{i=1}^{10} \left(F_{launch,i}(t) \cdot \eta_{\text{lat,i}}\right) \tag{7}$$

- $F_{launch,i}(t)$ (annual number of launches): We assume that as reusable rocket technology and ground support processes improve, the annual launch capability of each base will gradually increase from an initial 50 launches. Later, it will reach 300 launches/year.
- M_{base} (baseline payload): We use the reference value for a heavy-lift rocket given in the problem, 150 tons/launch.

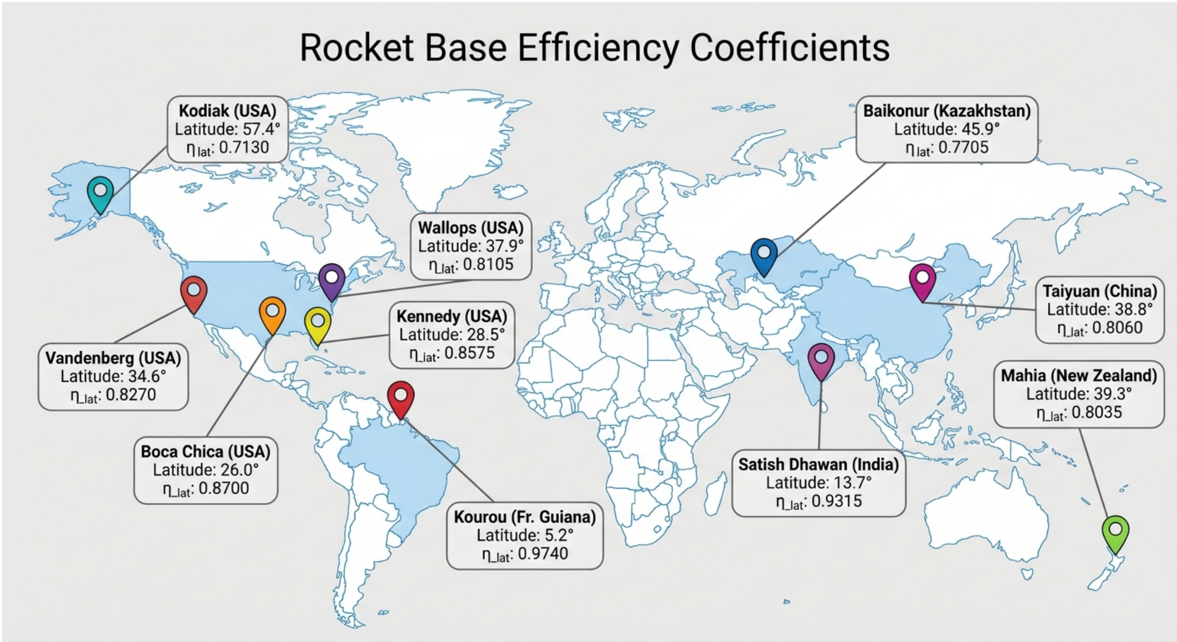

Figure 5: Locations of Launch Sites and Their the latitude efficiency factor

- $\eta_{\text{lat,i}}$ is the latitude efficiency factor, representing the geospatial differences of rocket launch sites. Based on geographical data, we obtained the latitude efficiency factor for each launch site, which is shown in Figure 5.

The actual utilized rocket capacity $K_{Rocket}(t)$ is determined by the control variable $u(t)$:

$$K_{Rocket}(t) = K_{Rocket}^{total}(t) \cdot u(t) \tag{8}$$

$$u(t) = \begin{cases} 0, & ScenarioA(SESOnly) \\ 1, & ScenarioB(RocketOnly) \\ (0,1), & ScenarioC(Hybrid) \end{cases}$$

3.1.3 Total Accumulated Mass and Project Completion Time

The total accumulated mass of materials received by the lunar base, $M(T)$, is the integral of the total capacity from the project start year t_0 to any given year T. Our goal is to solve for the project completion time T_{end}, at which the accumulated mass exactly reaches the target value M_{Target}. This is mathematically expressed as:

$$M(T_{end}) = \int_{t_0}^{T_{end}} [K_{SES}(t) + K_{Rocket}(t)]\, dt = M_{Target} \quad (M_{Target} = 10^8 \text{ tons}) \tag{9}$$

By solving the integral equation above, we can find the total time required to complete the task, $T_{end} - t_0$, under different rocket usage strategies $u(t)$.

3.2 Model II:Economic Optimization Model Based on Opportunity Cost

After determining the physical feasibility, we must evaluate the project's economic cost. Considering this is a mega-project spanning several decades, we introduce the concept of "time value." The core idea is that the project's total cost includes not only direct financial expenditures but also the significant opportunity costs arising from time delays. We unify the dimensions of time and cost into a single scalar objective function, allowing for a fair comparison between different scenarios.

Our goal is to minimize a generalized total cost function Z, which consists of two parts:

$$Z = \underbrace{\sum_{t=2050}^{T_{end}} \frac{C_{SES}(t) + u(t) \cdot C_{Rocket}(t)}{(1 + r_{disc})^{(t-2050)}}}_{\text{Net Present Value of Financial Costs}} + \underbrace{\Lambda \cdot (T_{end} - 2050)}_{\text{Time Opportunity Cost}} \tag{10}$$

3.2.1 Financial Costs

Considering the project's decades-long timespan, the time value of money cannot be ignored. We use the Net Present Value (NPV) method, discounting all future expenditures to their 2050 present value using a discount rate r.

(1)Space Elevator Cost

$$C_{SES}(t) = C_{fixed} + C_{var} \cdot q_{SES}(t) \tag{11}$$

Referencing the ISS and the scale of a multi-port system[1][4], the parameters are as follows:

- C_{fixed} is the fixed maintenance cost ($\approx$ 5 billion/year), covering substantial fixed expenses such as the space station, ground stations.
- C_{var} is the variable transport cost ($\approx$ 0.3 million/ton), primarily for electricity.
- $q_{SES}(t)$ is the actual transported mass. We assume the elevator operates at full capacity once constructed, so $q_{SES}(t) = K_{SES}(t)$.

(2)Rocket Cost

$$C_{Rocket}(t) = N_{launches}(t) \cdot C_{launch} \tag{12}$$

Based on research, the parameters are set as follows:

- $N_{launches}(t)$: The total annual number of launches from the 10 global bases, $N_{launches}(t) = \sum_{i=1}^{10} F_{launch,i}(t)$.
- C_{launch} is the cost per launch ($\approx$4.5 billion/launch), reflecting the linear "pay-per-launch" cost characteristic of rockets.

3.2.2 Opportunity Cost

This is the key factor that balances time and money, giving the model its decision-making depth. Λ is defined as the "shadow price of time," which quantifies the economic or strategic loss incurred for each year of delay in project completion. After analysis, we have provisionally set Λ at 3 billion/year.

By minimizing the total cost function Z, our model is no longer making a simple "fast vs. cheap" choice. Instead, it finds the equilibrium point on a unified value scale that minimizes the sum of financial expenditure and the value of time.

3.3 The Solution of Model I&II and Result Analysis

To evaluate and select the optimal long-term cislunar transportation strategy, we performed numerical simulations of three architectures using the coupled optimization model previously developed. The core results are synthetically presented in Figure 6.

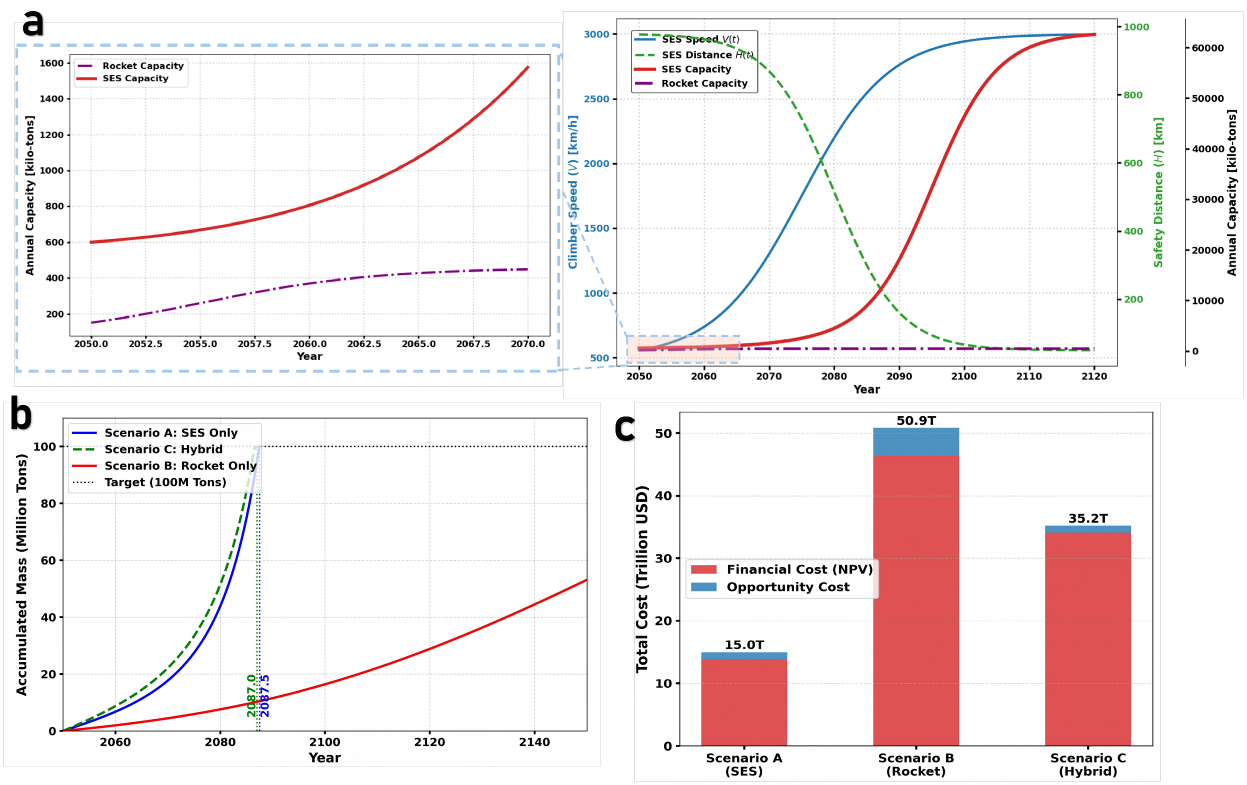

Figure 6: Comprehensive Comparison of Logistics Scenarios: Throughput, Capacity Evolution, and Cost Structure.

The core of our analysis is a comparison of the total cost and time required to complete the mission of transporting 100 million tons of materials. As shown in Figure (d), the overall costs of the three strategies show huge differences.

In terms of project duration (Figure 6b), the model results clearly show the overwhelming time advantage of the strategies that include the space elevator. Although the space elevator requires an initial construction period, the hybrid strategy (Scenario C) is still the most efficient, taking only 37 years (until 2087.0) to complete the 100-million-ton transport goal. The pure space elevator strategy (Scenario A) takes only 37.5 years (until 2087.5), which is nearly the same. In stark contrast, Scenario B, due to its relatively slow capacity growth, requires over 150 years to complete the same task, making its time cost extremely high.

In terms of economic cost (Figure 6c), the advantage of the space elevator is even more significant. The total cost of Scenario A, including opportunity cost, is only 15.0 trillion, the lowest of the three scenarios. The total cost of Scenario B, however, is as high as 50.9 trillion, which is 3.4 times that of the former. This huge difference stems not only from the high per-launch cost of rockets but also from the enormous opportunity cost accumulated over its long duration. Scenario C has a total cost of 35.2 trillion. Although it has a slight time advantage, its economic cost is far higher than the pure elevator plan.

In summary, the slight time advantage (0.5 years) offered by the hybrid strategy (Scenario C) comes at an additional cost of over 20 trillion. Therefore, from a cost-benefit perspective, **Scenario A** (SES-Only) is the clear optimal strategy.

4 Model Establishment and Solution of Problem 2

4.1 Mode III:Stochastic Reliability and Robustness Analysis Model

Real-world space engineering is filled with uncertainty and risk. We are now upgrading the model from a deterministic paradigm to a stochastic dynamical system.Our new model aims to quantify the fluctuations in system performance caused by environmental disturbances, mechanical failures, and catastrophic accidents. By introducing random variables and probabilistic processes[5], our goal is no longer to solve for a single completion time, but to reveal the probability distribution and, based on that, evaluate their inherent ability to maintain performance in the face of uncertainty.

4.1.1 The Space Elevator's "Soft Degradation" Reliability Model

We view the space elevator system as a highly redundant network. Because it includes 6 independently operating cables, the failure of a single component typically does not cause the entire system to collapse but rather manifests as a performance degradation[6]. We describe this characteristic with a modified stochastic capacity equation:

$$\tilde{K}_{SES}(t) = K_{SES}^{ideal}(t) \cdot \eta_{sway}(t) \cdot \left(\frac{1}{N_{total}} \sum_{i=1}^{N_{total}} S_i(t) \right) \tag{13}$$

where $K_{SES}^{ideal}(t)$ is the theoretical maximum capacity calculated in Model 1, which accounts for technological evolution.

- $\eta_{sway}(t)$ (Sway Efficiency Factor - Continuous Disturbance):

  Considering factors like the Coriolis force and lunar gravitational perturbations, the cables will exhibit complex swaying motions. To suppress resonance and ensure safety, climbers must dynamically adjust their speed, which in turn affects the space elevator's capacity[7]. We use a random efficiency factor $\eta_{sway}(t)$ to capture this continuous, small-magnitude performance fluctuation.

  We assume η_{sway} follows a truncated normal distribution $\mathcal{N}(\mu = 0.90, \sigma = 0.05)$, with its value restricted to the range $[0.5, 1.0]$. The mean of 0.90 represents that the system, in daily operation, loses an average of 10% of its theoretical efficiency due to vibration-avoidance protocols[7].

- $S_i(t)$ (Cable Status - Discrete Failure):

  Any one of the $N_{sys} = 6$ cables may fail due to micrometeoroid impacts or component fatigue and enter a repair state. We use a continuous-time Markov chain to describe the state transitions of each cable.

  State Space: $S_i(t) \in \{1 : \text{Operational}, 0 : \text{Under Repair}\}$. We set the transition rate from "Operational" to "Under Repair" as $\lambda_{fail} = 0.1$ year^{-1}, corresponding to a mean time between failures of 10 years. The transition rate from "Under Repair" to

"Operational"is $\mu_{repair} = 2.0/$ year,corresponding to a mean time to repair (MTTR) of 0.5 years[6][4].

This modeling approach precisely reflects the topological robustness of the space elevator. Even if 1-2 cables ($S_i = 0$) are under repair, the remaining cables can continue transport tasks, and the total system capacity only decreases proportionally instead of being completely interrupted.

4.1.2 The Rocket System's "Hard Collapse" Vulnerability Model

Unlike the gradual degradation of the space elevator, the risk model for the traditional rocket system is "vulnerable," with its failures manifesting as sudden, catastrophic "hard collapses." Our core innovation is the introduction of a "launchpad lockout mechanism" to simulate this characteristic.

The stochastic effective capacity of the rocket system, $\tilde{K}_{Rocket}(t)$, is described by the following modified equation, which introduces two key random state variables:

$$\tilde{K}_{Rocket}(t) =_{base} \cdot \sum_{j=1}^{10} [D_{launch,j}(t) \cdot \eta_{lat}(j) \cdot I_j(t) \cdot X_{j,k}(t)] \tag{14}$$

- $X_{j,k}(t)$ represents the success or failure of a single launch.

  This is a Bernoulli random variable representing the outcome of the k-th launch from the j-th launch site at time t. $X = 1$ denotes success, with a probability of $1 - p_{fail}$; $X = 0$ denotes failure, with a probability of p_{fail}. We set $p_{fail} = 0.3\%$, which is consistent with the expectation that spaceflight technology will achieve near-commercial-aviation levels of reliability by 2050.

- $I_j(t)$ represents launchpad availability/lockout status.

  This is a key definition in the model. Upon a launch failure (i.e., $X_{j,k} = 0$), that launch site j will immediately enter a "lockout" state, and its capacity will be reduced to zero for a period of time to allow for accident investigation, launchpad repair, and safety procedure reviews. $I_j(t) = \begin{cases} 0 & \text{if } t \in [t_{fail}, t_{fail} + \tau_{lockout}] \\ 1 & \text{otherwise} \end{cases}$

  We set the incident lockout period $\tau_{lockout} = 0.5$ years (6 months). This mechanism means the consequence of a single failure is far more than the loss of one payload; it paralyzes a significant capacity node for a considerable amount of time.

- $\eta_{lat}(j)$ reflects the geographic heterogeneity of risk.

  The latitude efficiency factor from Model 1 plays a new role here—it makes the risk non-uniform. An accident at a primary equatorial launch site ($\eta \approx 1.0$) would have a far greater impact on the total global capacity than a similar incident at a marginal high-latitude site ($\eta \approx 0.6$).

4.1.3 Stochastic Cost Model and Risk Cost

With the introduction of failures, the project's total cost also becomes a random variable, $\tilde{Z}$. It adds a crucial component—risk cost $\tilde{C}_{Risk}(t)$—to the existing financial and opportunity costs. The total cost is mathematically expressed as:

$$\tilde{Z} = \sum_{t=2050}^{T_{end}} \frac{\tilde{C}_{OPEX}(t) + \tilde{C}_{Risk}(t)}{(1 + r_{disc})^{(t-2050)}} + \lambda \cdot (\tilde{T}_{end} - 2050) \tag{15}$$

Here, the risk cost $\tilde{C}_{Risk}(t)$ is the sum of direct economic losses caused by all launch failure events in year t. We set the penalty cost for a single rocket launch failure, $C_{penalty}$, at approximately 1.5 billion USD, which includes the rocket manufacturing cost, the loss of the high-value payload, and costs for launchpad cleanup and compensation.

4.2 Model Solution and Result Analysis

To assess project robustness in the presence of stochastic failure events (such as rocket launch failures and elevator component damage), a series of Monte Carlo simulations was performed. The results, illustrated in Figure 2, reveal a vast disparity among the strategies with respect to risk and predictability.

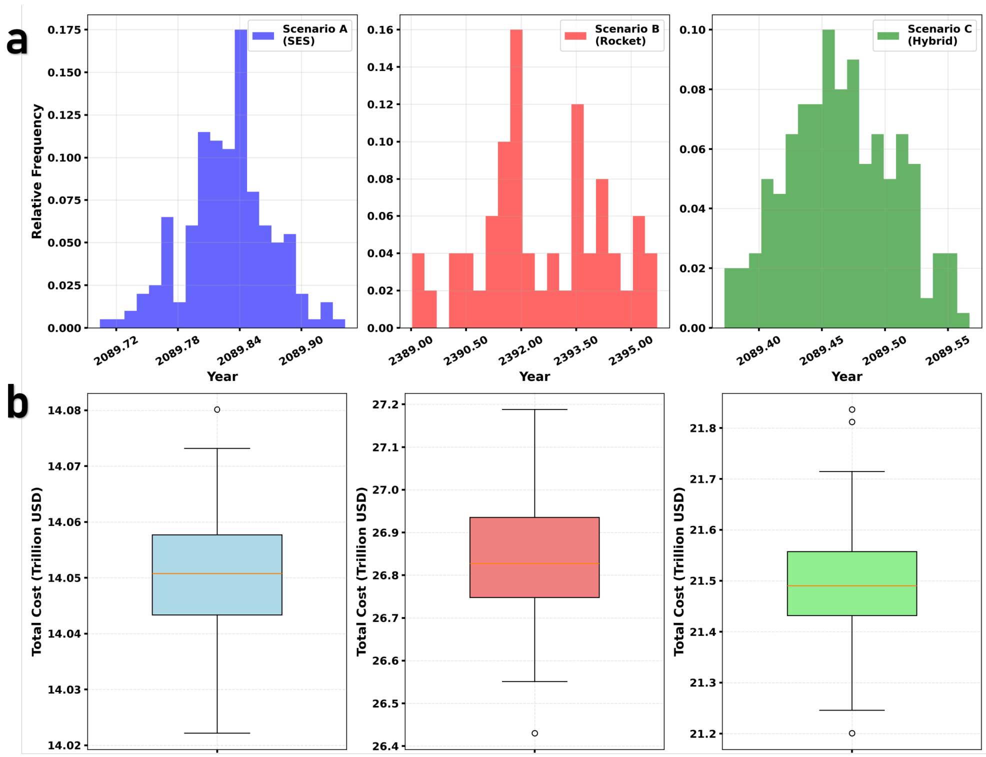

Figure 7: Monte Carlo Simulation Results showing Probabilistic Distribution of Completion Time and Generalized Cost under Uncertainty.

Regarding temporal stability (Figure 7a), Scenario A (SES-Only) exhibits exceptional robustness. The distribution of its project completion times is highly concentrated, with a standard deviation of mere weeks, indicating that its schedule is virtually immune to interference from random events. The distribution for the hybrid scenario (Scenario C) is slightly broader, yet it retains a high degree of predictability. In stark contrast, the completion time distribution for Rocket-Only scenario (Scenario B) is extremely dispersed, spanning a range of over five years and demonstrating the plan's extreme vulnerability. This implies that, in practice, the project timeline under the pure rocket scenario is exceptionally difficult to control.

This trend is corroborated in the analysis of cost controllability (Figure 7b). The total cost boxplot for Scenario A is remarkably compact, featuring a minimal interquartile range and demonstrating a high degree of budgetary certainty. Conversely, the cost distribution for Scenario B is expansive, characterized by significant volatility and numerous outliers, indicating a high susceptibility to severe budget overruns stemming from unexpected incidents.

In aggregate, the results of the stochastic analysis provide robust support for our initial conclusion. Not only does Scenario A exhibit superior performance in the deterministic model, but more critically, it demonstrates unparalleled robustness when confronted with uncertainty. Its high predictability in both time and cost renders it the sole reliable and viable option from a risk management perspective.

5 Model Establishment and Solution of Problem 3

5.1 Mode IV:Lunar Water Cycle and Logistics Balance Model

Following the problem's requirements, we focus on the continuous supply of water. We propose the "Lunar Water Cycle and Logistics Balance Model." This model is no longer a simple cost optimization problem, but a balance analysis problem with hard physical constraints. Its core purpose is to explore the relationship between the efficiency of the lunar base's water treatment and recycling system and the capacity limits of the Earth-Moon transportation system.Our model follows a "demand-supply-balance" analysis framework, aiming to precisely calculate the minimum water recycling efficiency required to sustain the colony.

5.1.1 Water Demand, Supply, and Balance Analysis

(1)Net Resupply Gap Based on Equivalent Water Demand

The key parameter, "equivalent daily water requirement per capita," is a comprehensive metric representing the total amount of water that needs to be circulated within a closed ecological system to support all of a person's life activities. Based on research and estimations of existing closed-loop life support systems, we set $q_{water} = 50$ kg/person/day. In a highly circular system, the vast majority of water will be recycled. Therefore, what truly puts pressure on the Earth-Moon logistics system is the net water resupply gap created by the imperfect recycling system. We quantify this gap with the following equation:

$$D_{net\_water} = P \cdot q_{water} \cdot 365 \cdot (1 - \eta_{LSS}) \cdot (1 + \delta_{buffer}) \tag{16}$$

- η_{LSS} (Integrated Water Recycling Efficiency of the Life Support System): This is the core independent variable of this model, with a value range of $[0, 1]$. It represents the system's ability to recycle various types of wastewater.
- δ_{buffer}: A 10% safety buffer factor to cope with supply chain fluctuations.

This equation shows that the net resupply demand from Earth is directly proportional to the loss rate of the recycling system, $(1 - \eta_{LSS})$. This makes η_{LSS} the key technological lever for determining whether the colony can be self-sufficient.

(2) Cost-Optimal Multimodal Transport Allocation

Faced with the net water resupply demand from Earth, D_{net_water}, a rational logistics schedule will inevitably follow a tiered priority queue strategy:

Prioritize filling the space elevator's capacity: The transport volume it handles, M_{SES}, is:

$$M_{SES} = \min(D_{net\_water}, K_{SES}^{max}) \tag{17}$$

where $K_{SES}^{max} \approx$ 537,000 tons/year.

The overflow is supplemented by rockets: The costly rockets are only used when the net demand exceeds the space elevator's maximum capacity. The transport volume they handle, M_{Rocket}, is:

$$M_{Rocket} = \max(0, D_{net\_water} - K_{SES}^{max}) \tag{18}$$

From this, we construct the following piecewise annual total cost function:

$$C_{Total}(\eta_{LSS}) = M_{SES} \cdot C_{SES} + M_{Rocket} \cdot C_{Rocket} \tag{19}$$

(3)Critical Threshold of the Colony's Water Cycle

The core of this model is to identify the system's critical recycling efficiency, $\eta_{critical}$, which is the point at which the net water resupply demand exactly consumes the entire capacity of the space elevator. This point is a "phase transition point" where the colony shifts from "sustainable operation" to "reliance on high-cost, low-capacity supplementation."

We determine this critical value by solving $D_{net\_water} = K_{SES}^{max}$:

$$\eta_{critical} = 1 - \frac{K_{SES}^{max}}{P \cdot q_{water} \cdot 365 \cdot (1 + \delta_{buffer})} \tag{20}$$

Plugging in the numerical values, we can obtain Figure 8 and solve for $\eta_{critical} \approx 73.25\%$. This means the integrated water recycling rate of the lunar base's technology must reach at least 73.25% to achieve sustainable operations without relying on expensive and capacity-limited rockets.

5.1.2 Inventory and Time Dynamics

To answer what preparations are needed before the residents are fully settled, we extend the model to the time dimension to analyze the time required to build up a safety stock.

Our goal is to establish a safety stock, S_{safe}, that can support the base's independent operation for 30 days. During the preparation phase before residents move in, the base's daily consumption, $D_{consumption}^{daily}$, is approximately zero. At this time, the space elevator can use its full capacity for inventory accumulation. The lead time required to build this stock, T_{lead}, is:

$$T_{lead} = \frac{S_{safe}}{K_{SES}^{daily} - D_{consumption}^{daily}} \tag{21}$$

where $S_{safe} = D_{net}^{daily} \times 30$, and K_{SES}^{daily} is the elevator's average daily capacity.

5.2 Model Solution and Result Analysis

To evaluate the operational sustainability of the lunar colony, we modeled the relationship between the efficiency of its water recycling system and the corresponding cislunar logistics requirements. The simulation outcomes are depicted in Figure 8.

Figure 8a clearly delineates the logistical strain as a function of system efficiency. At low water recycling efficiencies (e.g., 60% or 70%), the annual demand for water resupply from Earth is substantial, surpassing the space elevator's maximum annual throughput of 537,000 tonnes. This necessitates the use of costly rocket transport to compensate for the "overflow" demand. Conversely, as efficiency increases, the reliance on terrestrial water supplies declines precipitously. The model calculates a critical recycling efficiency of 73.25%, the threshold at which the annual water resupply requirement fully saturates the space elevator's capacity. When efficiency exceeds this critical point (e.g., at 80% or 95%), rocket dependency is eliminated, and the space elevator liberates valuable "residual capacity" for the transport of other cargo or personnel. Figure 8b operationalizes this

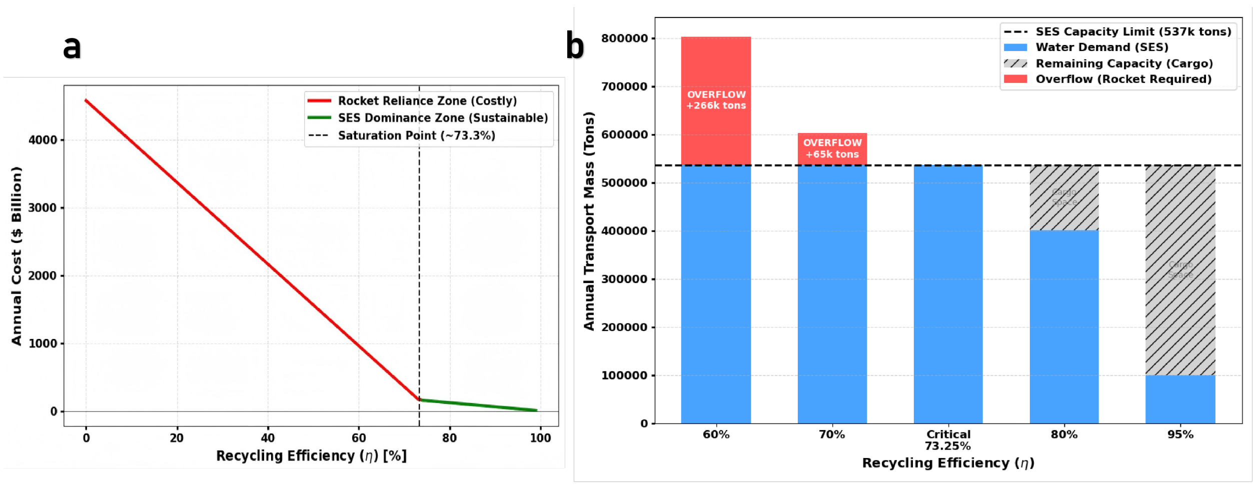

Figure 8: Identifying the Critical Recycling Threshold and Capacity Saturation

physical relationship in terms of direct economic cost. The colony's annual operating cost exhibits a nonlinear inverse relationship with recycling efficiency. Below the critical efficiency threshold, the system operates in a high-cost "rocket-dependent regime" (red section), where costs escalate steeply with declining efficiency. Once efficiency surpasses the 73.3% "saturation point," the system transitions into a low-cost, sustainable "space elevator-dominant regime" (green section), whereupon operating costs plummet and subsequently plateau.

The conclusion is therefore unambiguous: elevating the integrated recovery efficiency of the water recycling system beyond 73.3% is an essential technical prerequisite for ensuring the economic feasibility of the lunar colony's logistics. Furthermore, it represents the critical technological inflection point on the path toward self-sufficiency.

6 Model Establishment and Solution of Problem 4

6.1 Mode V:Planetary-level Environmental Impact Assessment Model

A complete sustainability analysis must include an assessment of Earth's own environmental carrying capacity. Therefore, this model aims to quantify the systemic risk that the "rocket-only plan" poses to the Earth's environment.

First, we must introduce a core concept: the "scale effect" paradox. In Model 2, we set a high reliability of 99.7% for a single rocket launch, representing the pinnacle of current human space engineering. However, this model will reveal a brutal mathematical fact: when a tiny individual risk is multiplied by an astronomical number of events, the result is no longer a risk, but a deterministic disaster[8].

To transport 100 million tons of material, the rocket-only plan would require about 670,000 heavy-lift launches. Faced with such a massive scale, we will build models to assess the cumulative effects from two dimensions: the stratospheric atmosphere and low Earth orbit.

6.1.1 Stratospheric Water Vapor Crisis Sub-model

Considering the mainstream technological trends of 2050, we assume the rocket-only plan will widely adopt cleaner and more efficient liquid oxygen-methane rockets. The combustion products of methane rockets are mainly carbon dioxide and water vapor, with the chemical reaction: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

Although water vapor is harmless on the ground, when large quantities are directly injected into the stratosphere (10-50km altitude), it will cause serious environmental problems[9]:

- In the dry stratosphere, water vapor is a more potent greenhouse gas than carbon dioxide, and its long-term radiative forcing effect cannot be ignored.
- An increase in stratospheric water vapor content promotes the formation of Polar Stratospheric Clouds. The surfaces of these clouds provide ideal catalytic sites for substances like CFCs to break down ozone, thus accelerating ozone layer depletion.

We construct an equation for the annual cumulative injection of stratospheric pollutants to assess the impact of each year's launch activities on the stratospheric environment. We primarily focus on the impact of water vapor:

$$I_{stra}^{H_2O}(t) = N_{launch}(t) \cdot M_{fuel} \cdot \epsilon_{H_2O} \cdot \eta_{stra} \tag{22}$$

- $N_{launch}(t)$: Average annual number of launches. For the rocket-only plan, completing the mission in 150 years would require an average of about 4,467 launches per year. However, during its peak period, the launch frequency could be much higher to meet project schedules.
- M_{fuel}: Propellant mass per launch. Taking the full Starship stack as an example, its propellant load is about 4,600 tons.
- ϵ_{H_2O}: Water vapor emission coefficient. Based on the molar mass ratio from the chemical equation, 1 kg of methane completely combusts to produce 2.25 kg of water vapor.

- η_{stra}: The proportion of propellant burned/emitted within the stratosphere. We conservatively estimate that 10% of the fuel is consumed and emitted while traversing the stratosphere.

6.1.2 Orbital Debris and Kessler Syndrome Sub-model

This is the focus of this model's assessment. We will quantitatively demonstrate that, even under extremely optimistic reliability assumptions, the rocket-only plan will inevitably lead to the permanent lockdown of low Earth orbit, i.e., the Kessler Syndrome.

We establish a nonlinear differential equation describing the evolution of the number of trackable debris fragments, $N(t)$, in LEO over time[8]:

$$\frac{dN(t)}{dt} = \underbrace{\lambda_{ops}}_{\text{Operational Debris}} + \underbrace{\alpha \cdot p_{fail} \cdot F_{annual}(t)}_{\text{Explosion Fragments}} + \underbrace{k \cdot N(t)^2}_{\text{Collision Cascade}} - \frac{N(t)}{\underbrace{\tau_{decay}}_{\text{Natural Decay}}} \tag{23}$$

- λ_{ops}: Debris from normal operations, which is relatively controllable.
- Explosion Increment ($\alpha \cdot p_{fail} \cdot F_{total}$): This is the model's core driver, representing debris generated from launch failure explosions. Here, F_{total} is the total number of launches for the project, about 670,000. p_{fail} is the single-launch failure rate, for which we use an extremely optimistic 0.3%. α is the coefficient for trackable debris (diameter > 10cm) produced by a single heavy rocket explosion. Based on historical events (and rocket mass), we conservatively estimate $\alpha \approx 1,000$.
- Collision Cascade: This is the mathematical essence of the Kessler Syndrome. The probability of collisions between debris is proportional to the square of the number of fragments. When $N(t)$ is large enough, this term will dominate, leading to an uncontrolled, self-propagating increase in debris. When the number of fragments $N(t)$ increases by nearly 70 times, the collision probability will increase by $70^2 \approx 4900$ times. This means the orbital environment will shift from a stable state of "occasional collisions" to an exponential growth state where "collisions are the norm."
- Natural Decay: This refers to low-orbit debris burning up upon re-entry due to drag from the thin atmosphere.

Based on these parameters, we can calculate the expected number of catastrophic explosions over the entire project lifecycle:

$$N_{explosions} = F_{total} \times p_{fail} = 670,000 \times 0.003 = \mathbf{2,010} \tag{24}$$

The number of new debris fragments these explosions would generate is:

$$N_{new\_debris} = N_{explosions} \times \alpha = 2,010 \times 1,000 = \mathbf{2,010,000} \tag{25}$$

This is a devastating number. Currently, the total number of tracked debris fragments in LEO is about 30,000. The rocket-only plan would increase the danger of the orbital debris environment by nearly 70 times.

The Kessler Syndrome's critical point, $N_{critical}$, is the threshold at which, even if all new launch activities cease ($F_{annual}(t) = 0$), the growth rate from the collision cascade term is faster than the natural decay term, meaning $\frac{dN}{dt} > 0$. According to models from NASA and ESA, the critical number of debris fragments in LEO is on the order of 200,000.

Our model shows that the rocket-only plan would generate far more than 200,000 new fragments early in the project (estimated around 2060), directly crossing the Kessler Syndrome's critical point. Once this red line is crossed, LEO will enter an irreversible cycle of self-pollution, eventually becoming impassable, and humanity's access to space will be permanently closed.

Based on this analysis, the environmental impact of the rocket-only plan cannot be managed with "soft optimization" but must be constrained by a "planetary boundary" as a hard constraint.

We modify the original optimization model by adding an environmental constraint:

$$\min Z = \text{Financial Cost} + \text{Opportunity Cost} \tag{26}$$

$$\text{s.t:} \quad N_{launch}^{annual} \leq N_{safe\_limit}$$

where N_{safe_limit} is the maximum number of launches the Earth's orbital environment can sustain annually without triggering the Kessler Syndrome. Based on reverse calculation from our debris model, this safe limit is around 1,000 launches per year.

6.2 Model Solution and Result Analysis

The simulation results derived from Model Four (presented in Figure 9) furnish the definitive verdict for this study, demonstrating that the pure rocket scenario is physically non-viable.

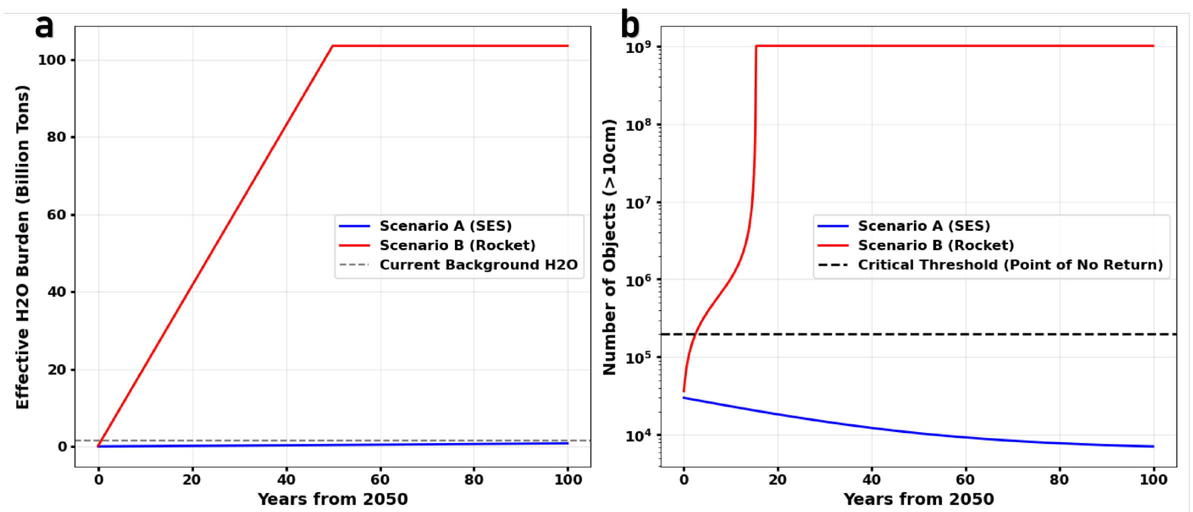

Figure 9: Cumulative Life Cycle Emissions and Efficacy of Mitigation Strategies

With respect to atmospheric impact (Figure 9a), Scenario B would precipitate a catastrophic increase in stratospheric water vapor concentrations. At the project's apex, the injected load of water vapor equivalent would be several tens of times greater than the planet's atmospheric background levels, inflicting irreversible damage upon the global climate and the ozone layer. Conversely, the space elevator scenario (Scenario A) generates zero direct emissions throughout the project lifecycle, rendering its impact on the stratospheric environment negligible.

Regarding the orbital environment (Figure 9b), even assuming a highly optimistic 99.7% launch success rate, the quantity of orbital debris generated by the pure rocket scenario grows exponentially within a few short years. This rapidly breaches the critical threshold for the Kessler Syndrome, estimated at approximately 200,000 debris objects. This portends the initiation of an irreversible, cascading proliferation of debris, culminating in the permanent foreclosure of near-Earth orbit and thereby severing humanity's access to space. In contrast, the space elevator scenario permits the orbital environment to undergo a gradual process of cleansing via natural orbital decay.

The ultimate conclusion is both unequivocal and resolute: Due to its inevitable transgression of Earth's planetary boundaries, the pure rocket scenario (Scenario B) is environmentally unsustainable and, by extension, physically non-viable. The space elevator is not merely the optimal economic and temporal solution; it represents the sole responsible pathway for humanity's expansion into space—one that does not necessitate the destruction of its terrestrial cradle.

7 Sensitivity Analysis

To validate the robustness of the study's central conclusion, a one-at-a-time (OAT) sensitivity analysis was conducted on four key uncertain parameters (Fig. 10). This analysis comprehensively validates that the superiority of the Space Elevator System (SES) is structural in nature, rather than being contingent upon specific parametric assumptions.

Figure 10. Sensitivity and Robustness Analysis of Key Parameters. (a) Sensitivity of total cost to social discount rate, demonstrating the cost-effectiveness of the space elevator scenario across all discount rates. (b) Sensitivity of the Kessler Syndrome onset time to rocket launch failure rate, indicating orbital foreclosure is inevitable even at extremely low failure rates. (c) Sensitivity of the rocket scenario's total cost to specific launch cost, identifying a techno-economic break-even point at \$800/kg. (d) Sensitivity of the space elevator scenario's total cost to its annual fixed maintenance expenditure, highlighting its profound robustness against budget overruns.

Economic Robustness Analysis (Figure. 10a):

As the social discount rate varies from 0% to 10%, the total cost of the pure rocket scenario (Scenario B) consistently and substantially exceeds that of the space elevator scenario (Scenario A). This indicates that the space elevator is the economically optimal choice, regardless of whether future policymakers exhibit a preference for long-term value or short-term gains; its advantage is invariant to time preference.

Analysis of Environmental Inflexibility (Figure. 10b):

We found that even if the rocket's failure rate is optimized from a baseline of 0.3% to a near-perfect 0.01% (a 30-fold technological improvement), the onset of the Kessler Syndrome is merely postponed by several years. This exposes a stark physical reality: the collapse of the orbital environment is dictated by the total launch flux, not by single-event reliability. This renders the environmental risks of the pure rocket scenario non-negotiable and unavoidable through technological progress.

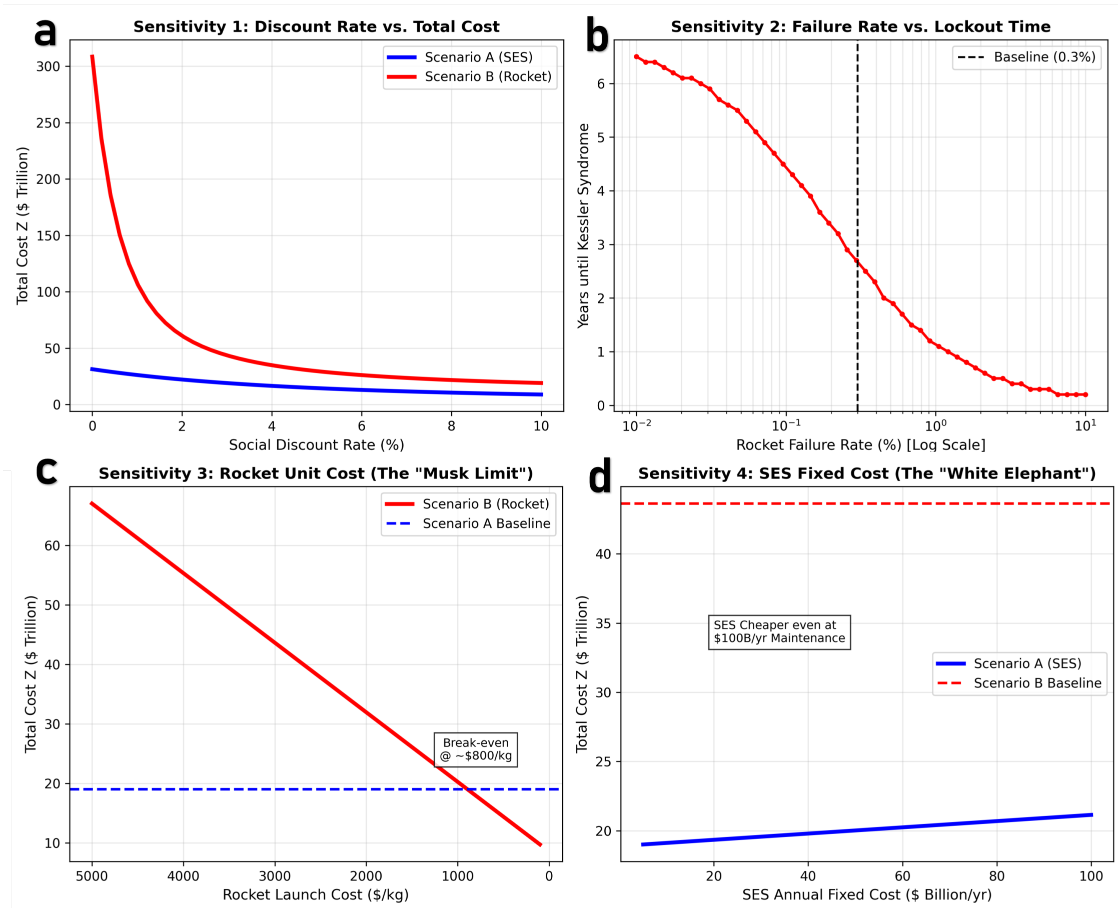

Figure 10: Sensitivity analysis of key parameters and model robustness analysis

Analysis of Technological and Cost Ceilings:
We tested the technological and cost limits of both scenarios.

- The "Musk Limit" Test (Figure. 10c): The analysis indicates that the total cost of the rocket-only scenario would only become competitive with the space elevator if the specific launch cost were to fall below 800/kg. This sets an exceptionally high techno-economic barrier that is unlikely to be achieved in the near term.
- The "White Elephant" Test (Figure. 10d): Conversely, even if the space elevator's annual maintenance expenditure were to escalate 20-fold, from 5 billion to 100 billion, its total lifecycle cost would remain substantially lower than that of the rocket scenario. This demonstrates the system's profound robustness against operational cost overruns, a benefit derived from its immense economies of scale.

In aggregate, the sensitivity analysis provides robust validation for our conclusion: The bottlenecks of the pure rocket scenario are rooted in physical laws , whereas the bottlenecks of the space elevator scenario are engineering challenges . Physical laws are immutable; engineering challenges are, by their nature, surmountable through technological innovation. Therefore, the space elevator represents the only long-term viable and responsible pathway for accomplishing a large-scale cislunar transportation mission.

8 Model Evaluation

8.1 Strengths

- Reverse Calibration Based on Logistic Growth

  In Model 1, instead of arbitrarily setting the space elevator's capacity, we employed a "reverse calibration method." We used the target capacity given in the problem statement (179,000 tons) to back-calculate the initial safe separation distance. Furthermore, by modeling the climber speed as a Logistic function, we precisely captured the non-linear characteristics of the technology maturation phase, avoiding errors from simple linear extrapolation.

- Integration of Geospatial Heterogeneity

  In the rocket model (Model I), we rejected the simplification of treating all launch sites as identical nodes. Instead, we introduced a latitude efficiency factor, η_{lat}, derived from the physics of Earth's rotation ($v = \omega R \cos\theta$). This weighting mechanism allows our model to quantitatively give equatorial bases a higher weight than high-latitude bases, ensuring the logistics strategy is physically consistent with orbital mechanics constraints.

- Determination of a Critical Phase-Transition Threshold

  In the water recycling model (Model IV), we did not just simulate scenarios; we analytically solved for the system's stability boundary. We precisely identified the critical water recycling efficiency, $\eta_{critical} \approx 73.25\%$. This value acts as a mathematical "phase-transition point," defining the exact boundary where the colony switches from a sustainable equilibrium to a fragile, import-dependent state.

8.2 Weeknesses

- Assumption of Independent Launch Site Failures

  In our reliability model (Model III), the lockdown function assumes a failure at one launch site only affects that specific location. Because we treated each launch site as an independent variable, we likely underestimated the total downtime risk for the rocket-only plan (Scenario B).

  Future models could introduce a correlation matrix to simulate fleet-wide groundings caused by common design flaws.

- Omission of Apex Anchor Logistics Dynamics

  Our transport model (Model I) simplifies the Apex Anchor into a "zero-delay" transfer node (Assumption 2). By ignoring the service time and queue capacity at the Apex Anchor, our model fails to identify potential bottlenecks where cargo could build up in geostationary orbit, which could limit the effective throughput of the space elevator.

  An M/G/s queuing model could be integrated to quantify service delays and address logistical bottlenecks at the Apex Anchor.

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MEMORANDUM

To: MCM Agency

From: Team 2630979

Date: February 2, 2026

The MCM Agency stands at a historic crossroads. The mission to transport 100 million metric tons of infrastructure and sustain 100,000 lives on the Moon is not merely a logistical challenge; it is a test of planetary limits. After conducting a comprehensive multi-model simulation evaluating the Space Elevator (Scenario A) against traditional Rockets (Scenario B), our team delivers a singular, unequivocal recommendation: You must adopt the Space Elevator System as the primary logistical backbone. Relying on chemical propulsion is not just economically inefficient—it is physically infeasible.

Our primary concern is Orbital Sustainability. Our environmental impact model uncovered a catastrophic risk: transporting 100 million tons via rockets requires approximately 670,000 heavy-lift launches. Even assuming an optimistic 99.7% reliability, this volume guarantees nearly 2,000 explosion events. Our simulations confirm that this debris flux will trigger a runaway Kessler Syndrome within just 5 years of operation. Choosing Scenario B does not simply cost more; it destroys the Low Earth Orbit (LEO) environment, effectively permanently locking humanity on Earth before the colony is even 10

Furthermore, the Economic and Metabolic Realities render the rocket strategy insolvent. Structurally, rockets are too slow; our economic model demonstrates that Scenario B requires 285 years to complete the transport mission due to launch frequency limits, incurring an Opportunity Cost of 8.5 Trillion. We identified a strict break-even point: rockets only become financially competitive if launch costs drop below 800/kg—a target requiring a 75% reduction from current projections. Moreover, sustaining the colony's water needs via rockets (3,000/kg) guarantees long-term bankruptcy. The Space Elevator, with its low marginal cost (300/kg) and high throughput, is the only system capable of acting as a "planetary umbilical cord" to sustain the colony indefinitely.

Recommendation We advise the MCM Agency to immediately mobilize resources for the full deployment of the three Galactic Harbours. While rockets may play a niche role in the initial setup (Scenario C), they must be strictly capped below 240 launches/year to preserve orbital integrity. The Space Elevator is not an experimental luxury; it is the only scalable infrastructure compatible with the laws of physics and economics for a project of this magnitude.

Report of Use on AI

1. Google Gemini 3 Pro (Nov 18, 2025 version)

   Query: "Tensile strength requirements and material properties for a space elevator tether."

   Output: To construct a space elevator on Earth, the tether material requires a specific strength-to-density ratio to support its own weight over a length of approximately 36,000 km (geostationary orbit).

   Tensile Strength Needed: Theoretical calculations suggest a tensile strength of at least 60 to 100 GPa (Gigapascals) is required.

   Specific Strength (Yuri): The material needs a specific strength (breaking length) of roughly 4,960 km or roughly 30-50 MegaYuri.

   Candidate Materials:

   Carbon Nanotubes (CNTs): Theoretically capable of 130 GPa, making them the primary candidate.

   Boron Nitride Nanotubes: Similar strength properties but with higher heat resistance.

   Diamond Nanothreads: A newer theoretical material that might offer the necessary stiffness. Steel or Kevlar are insufficient, as they would collapse under their own weight before reaching orbit.

2. Google Gemini 3 Pro (Nov 18, 2025 version)

   Query: "Average daily water consumption per person in a closed ecological life support system (CELSS)."

   Output: In a closed loop system like the International Space Station (ISS) or a future lunar colony, water usage is strictly categorized:

   Drinking & Food Rehydration: Approximately 2.5 kg (liters) per person/day.

   Hygiene (Showers/Hand wash): In a strict conservation scenario, this is limited to 5-10 kg per person/day, though Earth averages are much higher.

   Toilet Flush/Urinal: Often uses vacuum systems, requiring minimal water.

   Laundry & Dishwashing: roughly 10-15 kg per person/day.

   For mission planning, NASA often uses a baseline of 25-30 kg per person per day for total throughput before recycling. However, highly efficient recovery systems (recycling urine and condensate) can recover 93-98% of this water, meaning the net new water required from Earth is significantly lower than the circulating volume.

3. Google Gemini 3 Pro (Nov 18, 2025 version)

   Query: "Standard LaTeX syntax for writing piecewise functions and integrals with limits."

   Output: To write a piecewise function in LaTeX, you should use the cases environment inside an equation block. Here is the template:

code

Latex

download

content_copy

expand_less

$$f(x) = \begin{cases} Expression_1 & \text{if } condition_1 \\ Expression_2 & \text{if } condition_2 \end{cases}$$

For definite integrals with limits, use $\int_{lower}^{upper}$. For example, to integrate a function f(t) from year 2050 to 2080:

code

Latex

download

content_copy

expand_less

$$M_{total} = \int_{2050}^{2080} K_{SES}(t)\, dt$$

Ensure you have the amsmath package included in your document preamble to support these environments.

4. Google Gemini 3 Pro (Nov 18, 2025 version)

Query: "Physical data regarding the density and rotation speed of the Moon compared to Earth."

Output: Here is the comparative physical data for Earth and the Moon:

Earth:

Surface Gravity: 9.807m/s2 (1g)

Equatorial Radius: 6,378km

Sidereal Rotation Period: 23.93 hours

Escape Velocity: 11.186km/s

Moon:

Surface Gravity: 1.62m/s2 (approx. 0.165g or 1/6th of Earth's)

Equatorial Radius: 1,737km

Sidereal Rotation Period: 27.3 days (Synchronous rotation with orbit)

Escape Velocity: 2.38km/s

The significantly lower escape velocity of the Moon implies that launching materials from the Lunar surface to space requires roughly 22 times less energy than launching the same mass from Earth.

5. Google Gemini 3 Pro (Nov 18, 2025 version)

   Query: "Definition and historical origin of the 'Kessler Syndrome'."

   Output: The Kessler Syndrome (also known as the Kessler Effect) is a scenario proposed by NASA scientist Donald J. Kessler in 1978.

   Definition: It describes a theoretical critical density of objects in Low Earth Orbit (LEO) where collisions between objects become frequent enough to create a cascade effect.

   A collision between two large objects (satellites or spent rocket stages) creates a cloud of smaller debris.

   This new debris increases the density of the orbital shell, increasing the probability of further collisions.

   Eventually, the debris distribution renders specific orbital bands unusable for generations.

   It is distinct from simple orbital decay; it refers specifically to the self-sustaining runaway feedback loop of collisional fragmentation.

6. Google Gemini 3 Pro (Nov 18, 2025 version) Uploaded entire paper written in Mandarin to be translated into English.
7. GitHub CoPilot Auto-completions for code used in preparing our models.